# Multi Agent System (MAS) for Crypto Portfolio Management

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# Project Vision

Building a fully autonomous, self-improving, memory-driven, multi-agent intelligence system capable of performing

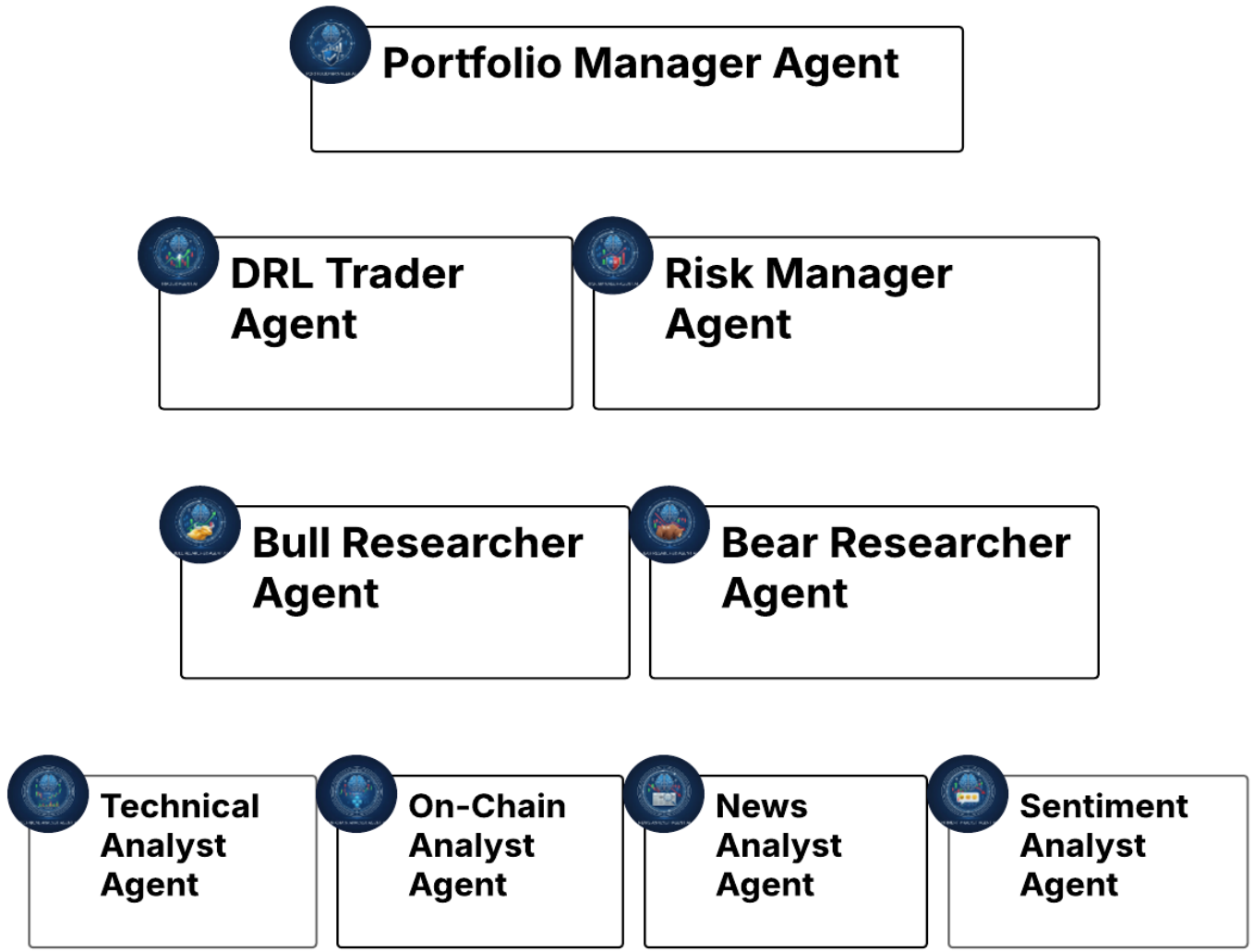
complete crypto market research,

debating investment decisions,

managing risk,

and generating final trade actions.

# MAS Architecture



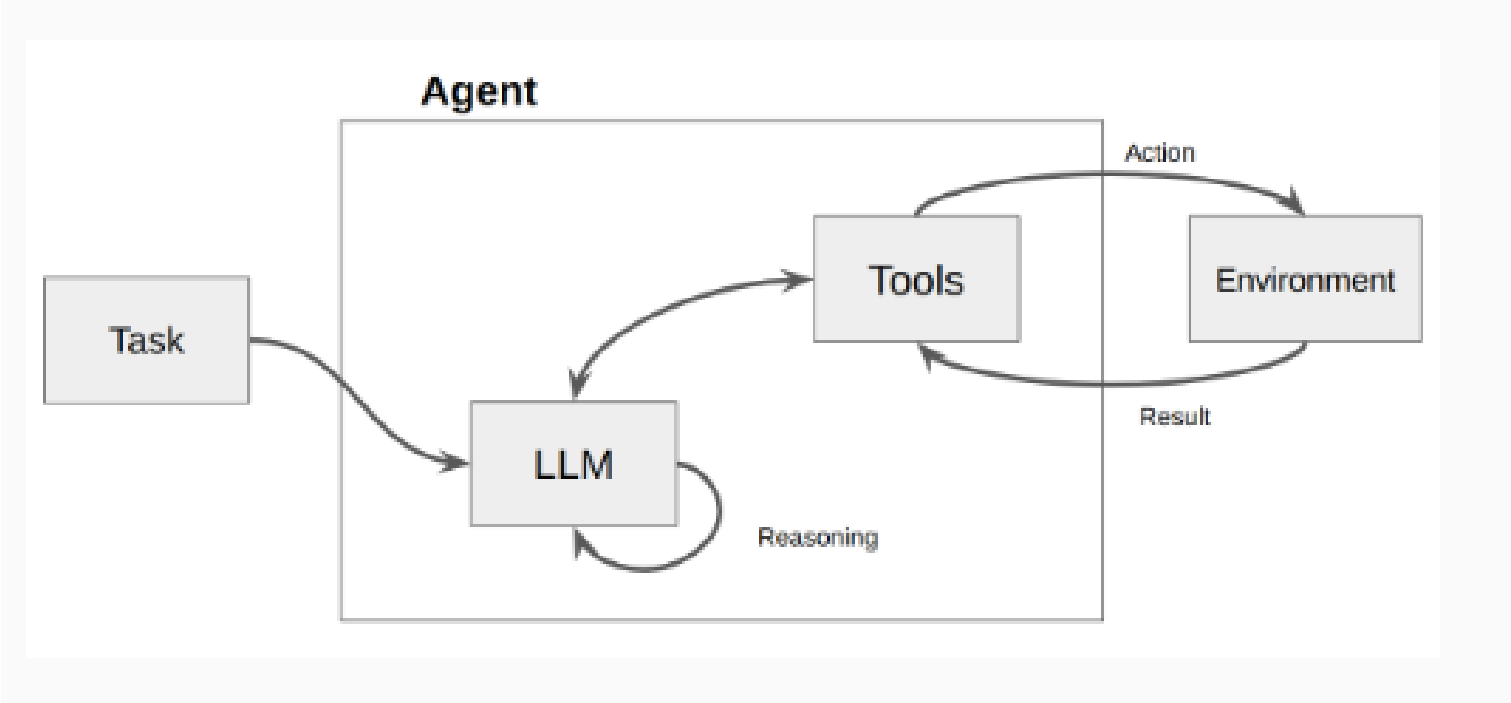
# Multi-Agent Orchestration and Collaboration Framework

ReAct Framework

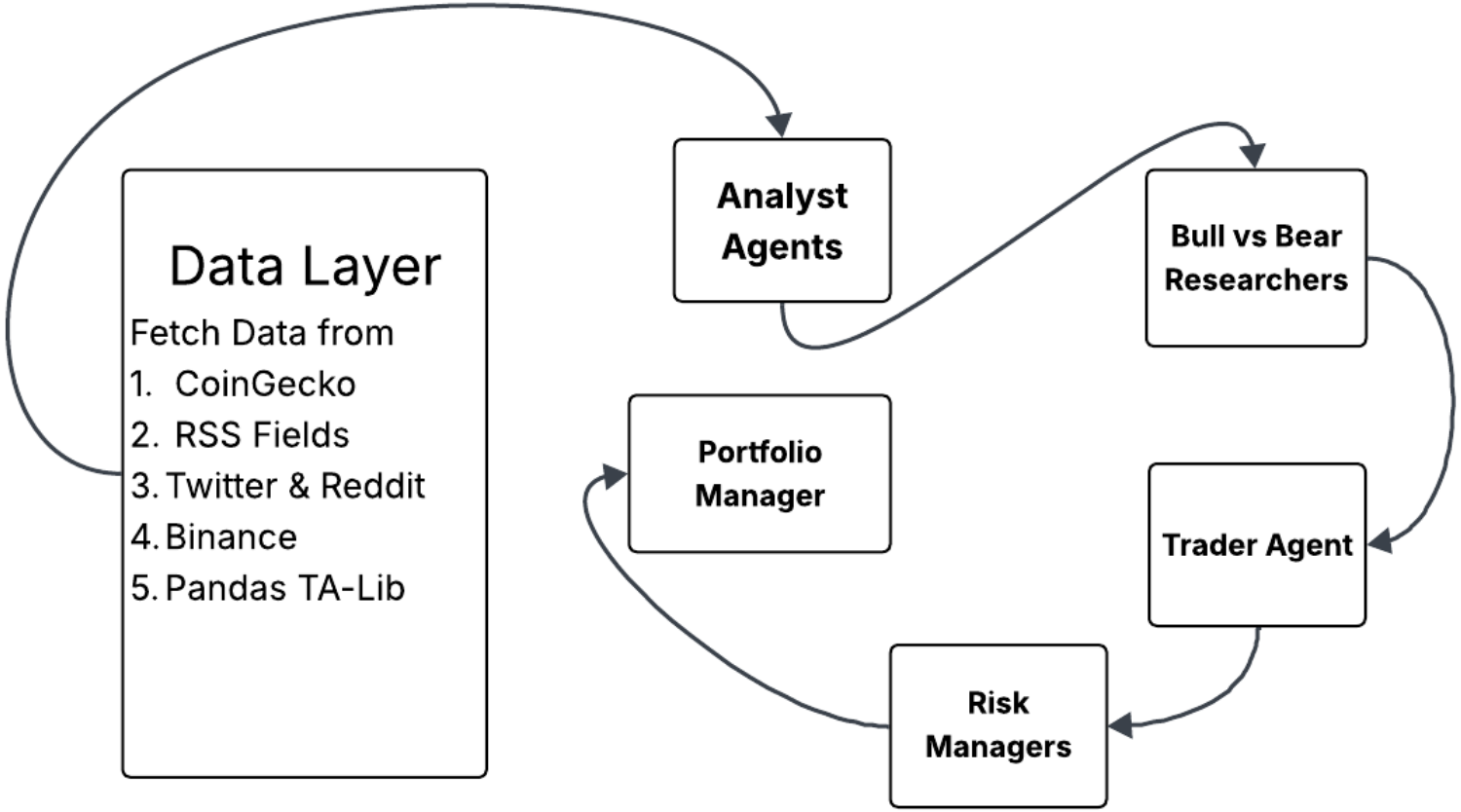
LangGraph

LangChain

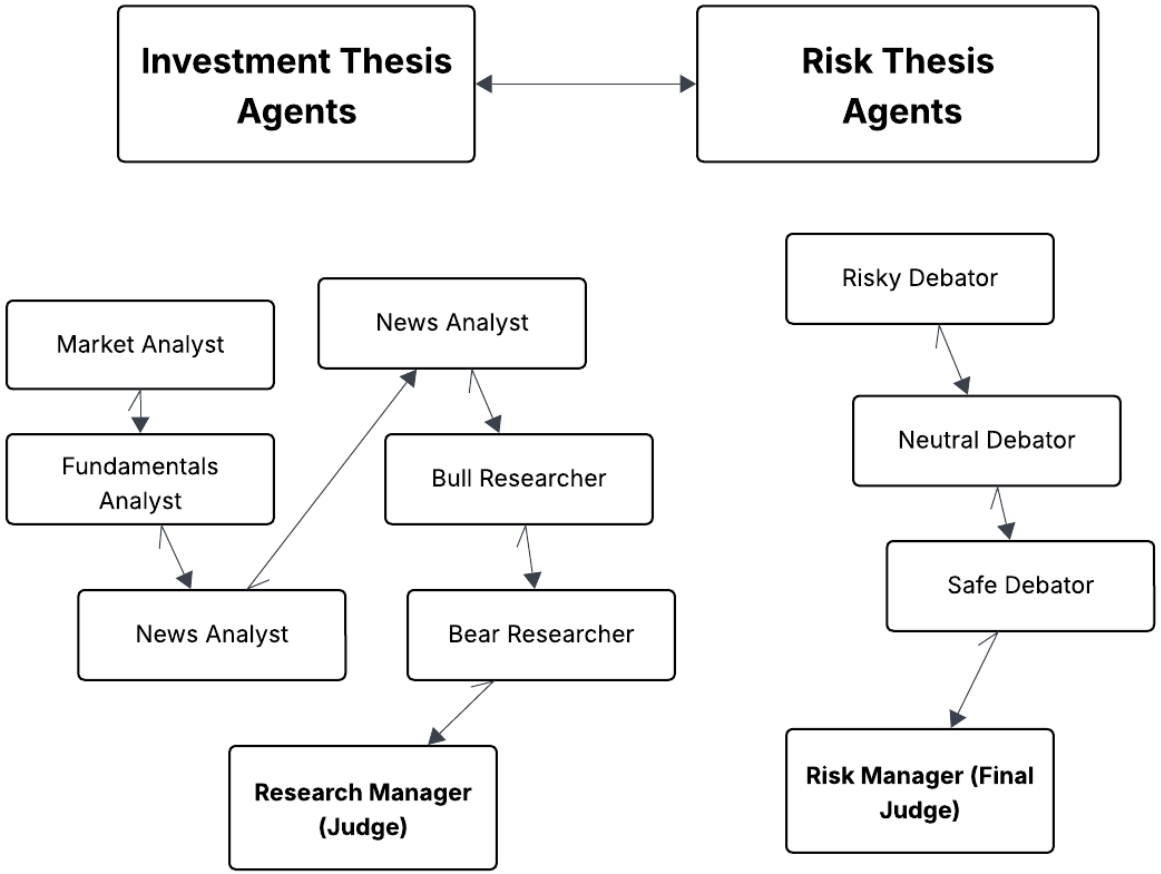
# ReAct Framwork



# Multi-Agent Workflow



# Real LangGraph Agents Workflow



# User Interface Layer

# Outputs Generated via **CLI** and **Limitations**

**Each run produces:**

Final decision

Debate transcripts

Memory updates

**For presentation clarity:**

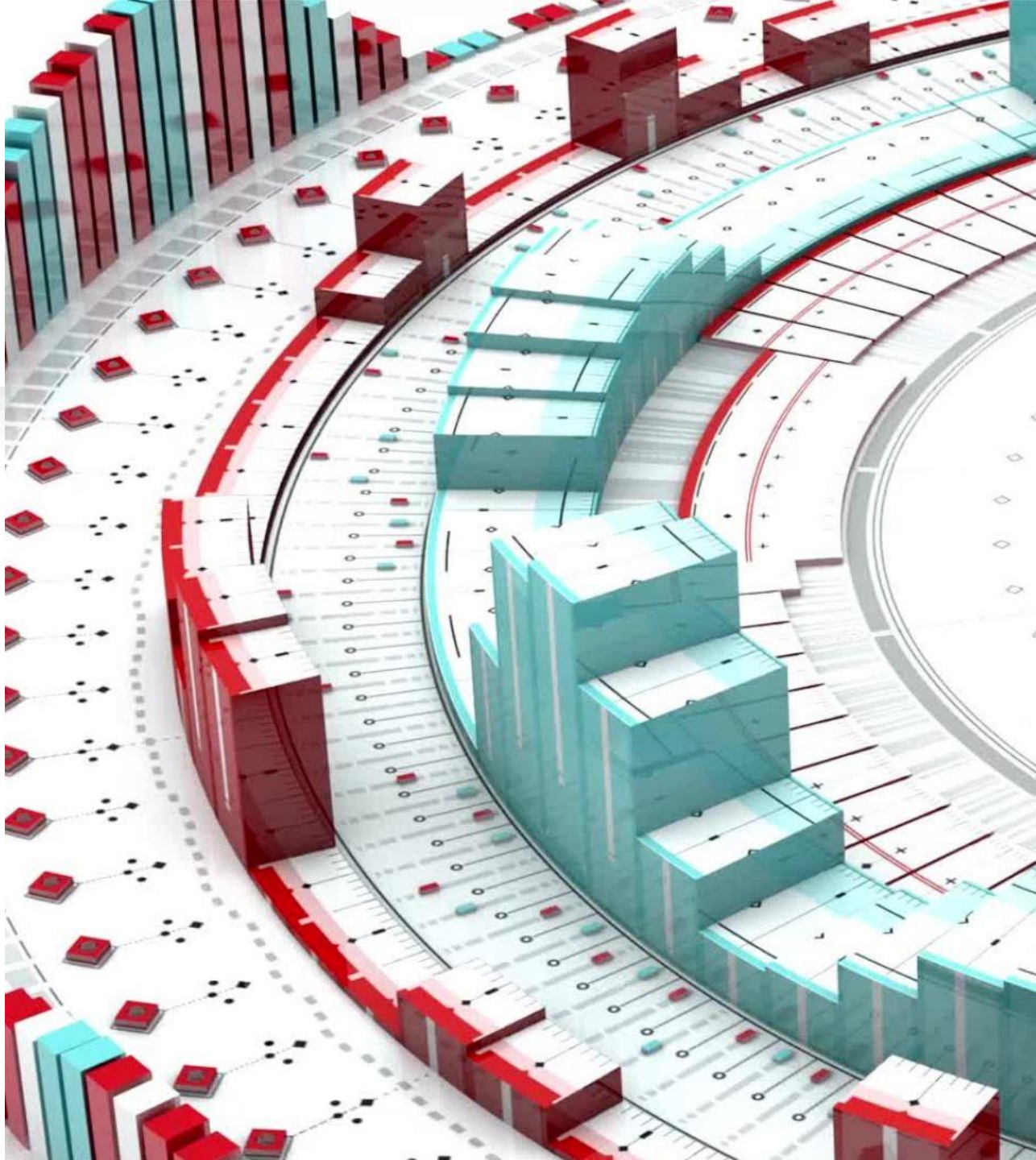
Results are summarized instead of raw logs

**Current limitations include:**

API rate limits (Gemini ops)

# SlashAgents Output Time

1. This is the real-time data and LLM report extraction engine of my AI system.
2. Here you can see the system
3. pulling live crypto prices,
4. building OHLCV datasets,
5. generating technical indicators,
6. and extracting market fundamentals.
7. This pipeline is the backbone of the entire intelligent trading ecosystem.



# Crypto Price Data (OHLCV)

| Date | Open (USDT) | High (USDT) | Low (USDT) | Close (USDT) | Volume (BTC) |
|---|---|---|---|---|---|
| 2025-01-01 | $93,576.00 | $95,151.15 | $92,888.00 | $94,591.79 | 10,373.33 |
| 2025-01-02 | $94,591.78 | $97,839.50 | $94,392.00 | $96,984.79 | 21,970.49 |
| 2025-01-03 | $96,984.79 | $98,976.91 | $96,100.01 | $98,174.18 | 15,253.83 |
| 2025-01-04 | $98,174.17 | $98,778.43 | $97,514.79 | $98,220.50 | 8,990.06 |
| 2025-01-05 | $98,220.51 | $98,836.85 | $97,276.79 | $98,363.61 | 8,095.64 |
| 2025-01-06 | $98,363.61 | $102,480.00 | $97,920.00 | $102,235.60 | 25,263.43 |
| 2025-01-07 | $102,235.60 | $102,724.38 | $96,181.81 | $96,954.61 | 32,059.88 |
| 2025-01-08 | $96,954.60 | $97,268.65 | $92,500.90 | $95,060.61 | 33,704.68 |
| 2025-01-09 | $95,060.61 | $95,382.32 | $91,203.67 | $92,552.49 | 34,544.84 |
| 2025-01-10 | $92,552.49 | $95,836.00 | $92,206.02 | $94,726.11 | 31,482.86 |
| 2025-01-11 | $94,726.10 | $95,050.94 | $93,831.73 | $94,599.99 | 7,047.90 |
| 2025-01-12 | $94,599.99 | $95,450.10 | $93,711.19 | $94,545.06 | 8,606.87 |
| 2025-01-13 | $94,545.07 | $95,940.00 | $89,256.69 | $94,536.10 | 42,619.56 |
| 2025-01-14 | $94,536.11 | $97,371.00 | $94,346.22 | $96,560.86 | 27,846.62 |
| 2025-01-15 | $96,560.85 | $100,681.94 | $96,500.00 | $100,497.35 | 30,510.00 |

# Bitcoin (BTC) Key Market Overview

```
======================================================
CRYPTO FUNDAMENTAL ANALYSIS: Bitcoin (BTC)
======================================================

MARKET DATA
-----------
Current Price:          $92.12K
Market Cap:             $1.84T
Market Cap Rank:        #1
Fully Diluted Value:    $1.84T
24h Trading Volume:     $82.75B

PRICE CHANGES
-------------
24h Change:             0.87%
7d Change:              -10.93%
30d Change:             -14.66%
1y Change:              -0.29%

SUPPLY (TOKENOMICS)
-------------------
Circulating Supply:     19,950,600
Total Supply:           19,950,600
Max Supply:             21000000.0
Supply %:               100.0%

ALL-TIME HIGH/LOW
-----------------
ATH:                    $126.08K
ATH Date:               2025-10-06
Distance from ATH:      -26.89%

ATL:                    $67.81
ATL Date:               2013-07-06
Distance from ATL:      135840.82%
```

```
==========================================
MARKET ACTIVITY (30 Days): BTC
==========================================

PRICE METRICS
-------------
Current Price:          $92.12K
30-Day Average:         $104.05K
30-Day Change:          -15.08%
Highest:                $114.48K
Lowest:                 $91.36K
Volatility:             22.21%

VOLUME METRICS (Activity Proxy)
-------------------------------
Recent Volume:          $82.75B
30-Day Avg Volume:      $69.97B
Volume Change:          -21.82%
Volume Trend:           Decreasing

LIQUIDITY FLOW
--------------
Avg Daily Turnover:     $69.97B
Peak Volume:            $113.20B
Low Volume:             $23.21B
```

```
========================================
MARKET PERFORMANCE: Bitcoin (BTC)
========================================

RETURNS (Revenue Equivalent)
-----------------------------
24h Return:                0.88%
7d Return:                 -10.91%
14d Return:                -10.62%
30d Return:                -14.64%
60d Return:                -20.40%
200d Return:               -3.94%
1y Return:                 -0.28%

MARKET POSITION (Market Share)
------------------------------
Market Cap:                $1.84T
Market Cap Rank:           #1

TRADING VOLUME (Activity/Engagement)
------------------------------------
24h Volume:                $82.71B
Volume/Market Cap:         0.0450

VALUATION METRICS
-----------------
Current Price:             $92.13K
High 24h:                  $92.94K
Low 24h:                   $88.54K

GROWTH INDICATORS
-----------------
ATH Performance:           -26.89% from ATH
ATL Performance:           135840.82% from ATL
Price vs FDV:              100.0%
```

# *BTC Market Performance*

# Technical Indicator Output

```
(.venv) C:\Users\Human\Documents\MAS\SlashAgents\slashagents>python -m dataflows.cr
fatal: bad revision 'HEAD'
================= TESTING LIST OF INDICATORS =================

Available indicators:
sma_50                - 50 SMA
sma_200               - 200 SMA
ema_10                - 10 EMA
ema_21                - 21 EMA
rsi                   - Relative Strength Index
macd                  - MACD Line
macd_signal           - MACD Signal Line
macd_histogram        - MACD Histogram
bbands_upper          - Bollinger Upper Band
bbands_middle         - Bollinger Middle Band
bbands_lower          - Bollinger Lower Band
atr                   - Average True Range
adx                   - Average Directional Index


================= TESTING INDIVIDUAL INDICATORS =================

TESTING RSI...
INFO:__main__:Calculating rsi for BTC
INFO:dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-01-01 to 2025-02-01
INFO:dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:dataflows.crypto_market_data:Successfully fetched 32 candles for BTC

======================================================================
RSI for BTC (2025-01-01 → 2025-02-01)
======================================================================
```

# Aggregate News Output

```
(.venv) C:\Users\Human\Documents\MAS\SlashAgents\slashagents\dataflows>python crypto_news.py

===== TEST 1: GLOBAL NEWS =====
INFO:__main__:Aggregating global crypto news...
INFO:__main__:Fetching RSS: coindesk
INFO:__main__:Fetching RSS: cointelegraph
INFO:__main__:Fetching RSS: decrypt
INFO:__main__:Fetching RSS: bitcoinmagazine
INFO:__main__:Fetching RSS: blockworks
INFO:__main__:Fetching RSS: utoday
Found 185 Articles
  [coindesk]  BlackRock Takes First Step Toward a Staked Ether ETF
  [coindesk]  Fed December Rate Cut Odds Collapse to 30%
  [coindesk]  Privacy-Focused Aztec Network's Ignition Chain Lights Up on Ethereum
  [coindesk]  India's Debt-Backed ARC Token Eyes Tentative Q1 2026 Debut, Sources Say
  [coindesk]  Dogecoin Hits Multi-Month Lows as Exchange Flows Turn Bullish for First Time in 6 Months
  [coindesk]  XRP Slumps as $2.15 Level Collapses, Bearish Structure Deepens
  [coindesk]  Asia Morning Briefing: Market Turns Defensive as Bitcoin Loses Its Bid
  [coindesk]  Stablecoin Spending Goes Mainstream With Opera MiniPay’s LatAm Integration
  [coindesk]  Samourai Wallet Co-Founder Bill Hill Sentenced to 4 Years in Prison for Unlicensed Money Transmitting
  [coindesk]  Nvidia Earnings Beat, Strong Outlook Calm Jittery Markets; Bitcoin Re-Takes $90K
  [coindesk]  DeFi Giant Spark Shelves Crypto App Plans to Focus on Institutional Infrastructure
  [coindesk]  Trump's Pick to Run CFTC, Selig, Tells Senators Crypto a 'Critical Mission' at Agency
  [coindesk]  Fed Rate-Cut Odds Plunge Further on Jobs Data Delays
  [coindesk]  New Hampshire Awaits Bitcoin Bond Buyer to Get First State Effort Rolling
  [coindesk]  Crypto Leverage Hits Record High in Q3 as DeFi Dominance Reshapes Market Structure: Galaxy
  [coindesk]  AI Agents Need Identity and Zero-Knowledge Proofs Are the Solution
  [coindesk]  Bitcoin Slips Back Below $90K – Crypto Correction Now Ranks Among Worst Since 2017, K33 Says
  [coindesk]  Stella's XLM Token Breaks Key $0.25 Support as Altcoins Suffer Continued Drawdown
  [coindesk]  Coinbase Debuts DEX Trading in Brazil as ‘Everything App’ Vision Grows
  [coindesk]  HBAR Slides 0.5% to $0.146 as Technical Support Crumbles
  [coindesk]  ICP Softens as Failed Breakout Above $5.17 Shifts Market Back Into Consolidation
  [coindesk]  BONK Extends Slide as Key Support Break Raises Prospect of Further Downside
  [coindesk]  Crypto Long & Short: Licences, Liquidity and the Shifting Geography of Exchange Quality
  [coindesk]  Senate Banking Panel Advances FDIC's Travis Hill for Wider Confirmation Vote
  [coindesk]  The Protocol: Hyperliquid Introduces Proposal to Cut Fees
  [cointelegraph]  Cathie Wood’s ARK loads up on Circle, BitMine, Bullish as crypto stocks slide
  [cointelegraph]  Bitwise XRP ETF to launch Thursday, but community questions ticker
  [cointelegraph]  Bitcoin’s drawdown shouldn’t be blamed on US shutdown or AI: Analysts
  [cointelegraph]  1 in 3 young investors switched advisers over crypto access: Survey
  [cointelegraph]  Cloudflare outage shows why crypto needs end-to-end decentralization
  [cointelegraph]  Bitcoin whale activity on track for its biggest week this year: Analysts
  [cointelegraph]  Warning: WhatsApp worm targets Brazilian crypto wallets, bank accounts
  [cointelegraph]  Abu Dhabi Investment Council triples stake in Bitcoin ETF in Q3: Report
  [cointelegraph]  Crypto, tech stocks climb on Nvidia’s blowout Q3 earnings
```

```
--- Turn 1 ---
⚒⚒ Analyst is calling: ['get_crypto_ohlcv']
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_ohlcv
INFO:slashagents.dataflows.interface:Category: market_data, Implementation: get_crypto_ohlcv
INFO:slashagents.dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-09-15 to 2025-12-15
INFO:slashagents.dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:slashagents.dataflows.crypto_market_data:Successfully fetched 92 candles for BTC
INFO:slashagents.dataflows.interface:Successfully executed get_crypto_ohlcv
```

# Analyst Agents Reporting

**On-chain Fundamentals Analyzing**

# Technical Indicators Analyzing

```
--- Turn 2 ---
□□ Analyst is calling: ['get_crypto_indicator', 'get_crypto_indicator', 'get_crypto_indicator', 'ge t_crypto_indicator', 'get_crypto_indicato
 'get_crypto_indicator', 'get_crypto_indicator', 'get_crypto_indicator', 'get_crypto_indicator', 'get_crypto_indicator']
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_indicator
INFO:slashagents.dataflows.interface:Category: technical_indicators, Implementation: get_crypto_indicator
INFO:slashagents.dataflows.crypto_technical_indicators:Calculating ema_10 for BTC
INFO:slashagents.dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-09-15 to 2025-12-15
INFO:slashagents.dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:slashagents.dataflows.crypto_market_data:Successfully fetched 92 candles for BTC
INFO:slashagents.dataflows.interface:Successfully executed get_crypto_indicator
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_indicator
INFO:slashagents.dataflows.interface:Category: technical_indicators, Implementation: get_crypto_indicator
INFO:slashagents.dataflows.crypto_technical_indicators:Calculating ema_21 for BTC
INFO:slashagents.dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-09-15 to 2025-12-15
INFO:slashagents.dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:slashagents.dataflows.crypto_market_data:Successfully fetched 92 candles for BTC
INFO:slashagents.dataflows.interface:Successfully executed get_crypto_indicator
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_indicator
INFO:slashagents.dataflows.interface:Category: technical_indicators, Implementation: get_crypto_indicator
INFO:slashagents.dataflows.crypto_technical_indicators:Calculating sma_200 for BTC
INFO:slashagents.dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-09-15 to 2025-12-15
INFO:slashagents.dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:slashagents.dataflows.crypto_market_data:Successfully fetched 92 candles for BTC
INFO:slashagents.dataflows.interface:Successfully executed get_crypto_indicator
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_indicator
INFO:slashagents.dataflows.interface:Category: technical_indicators, Implementation: get_crypto_indicator
INFO:slashagents.dataflows.crypto_technical_indicators:Calculating rsi for BTC
INFO:slashagents.dataflows.crypto_market_data:Fetching BTC OHLCV from 2025-09-15 to 2025-12-15
INFO:slashagents.dataflows.crypto_market_data:Fetching from binance: BTC/USDT (1d)
INFO:slashagents.dataflows.crypto_market_data:Successfully fetched 92 candles for BTC
INFO:slashagents.dataflows.interface:Successfully executed get_crypto_indicator
INFO:slashagents.dataflows.interface:Routing request for method: get_crypto_indicator
INFO:slashagents.dataflows.interface:Category: technical_indicators, Implementation: get_crypto_indicator
```

```
--- Turn 3 ---

📊 FINAL MARKET ANALYST REPORT:

[{'type': 'text', 'text': "FINAL TRANSACTION PROPOSAL: **SELL**\n\n**1. Indicator Selection Rationale:**\n\n*   **ema_10 & ema_21:** These short-term exponential moving
 averages are chosen to identify immediate trend direction and potential short-term crossovers, which can signal shifts in momentum. They are more responsive to recent
price changes than simple moving averages.\n*   **rsi:** The Relative Strength Index is a momentum oscillator that measures the speed and change of price movements. It
is used to identify overbought (>70) or oversold (<30) conditions and potential bullish/bearish divergences, signaling possible reversals.\n*   **macd, macd_signal, mac
d_histogram:** The Moving Average Convergence Divergence is a trend-following momentum indicator that shows the relationship between two exponential moving averages of
a security's price. The MACD line, signal line, and histogram provide insights into trend direction, strength, and potential reversals through crossovers and divergence
s.\n*   **bbands_upper, bbands_middle, bbands_lower:** Bollinger Bands measure market volatility and identify potential overbought or oversold conditions relative to th
e moving average. The bands expand and contract with volatility, and price touching or breaking a band can signal a continuation or reversal. The middle band acts as a
dynamic support/resistance.\n*   **adx:** The Average Directional Index is used to quantify trend strength. An ADX value above 25 generally indicates a strong trend, wh
ile values below 20 suggest a weak or ranging market. This helps confirm the reliability of other trend-following signals.\n\n**2. Detailed Technical Analysis for each
indicator (as of 2025-12-15):**\n\n*   **EMA 10 (89714.0057) & EMA 21 (90793.6387):**\n    *   The 10-day EMA (89714.01) is currently below the 21-day EMA (90793.64). T
his is a bearish crossover, indicating that the short-term momentum has shifted downwards.\n    *   Both EMAs are declining, suggesting a sustained short-term downtrend
.\n    *   The current price (86432.08) is significantly below both EMAs, confirming strong bearish momentum in the short term.\n    *   The EMA 21 (90793.64) is acting
 as a short-term resistance level.\n\n*   **RSI (37.0977):**\n    *   The RSI value of 37.10 is approaching the oversold territory (<30). While not yet oversold, it sug
gests that selling pressure has been significant recently.\n    *   A declining RSI from previous values (e.g., 49.53 on 2025-12-11) indicates increasing bearish moment
um.\n    *   There are no clear bullish divergences observed in the provided data.\n\n*   **MACD (-1604.9806), MACD Signal (-1734.1794), MACD Histogram (129.1988):**\n
   *   The MACD line (-1604.98) is above the MACD Signal line (-1734.18). This is a bullish crossover, which is a positive sign for short-term momentum, although the va
lues are still negative, indicating an overall bearish trend.\n    *   The MACD Histogram is positive (129.20), which confirms the bullish momentum from the MACD line c
rossing above the signal line. This suggests a potential reversal or at least a slowing of the downtrend.\n    *   However, looking at the recent trend, the MACD line h
as been consistently negative, indicating that the long-term momentum remains bearish. The recent positive histogram and crossover could be a temporary bounce within a
larger downtrend.\n\n*   **Bollinger Bands (Upper: 94090.1997, Middle: 90442.9860, Lower: 86795.7723):**\n    *   The current price (86432.08) is below the lower Bollin
ger Band (86795.77). This suggests that BTC is currently oversold on a short-term basis and could be due for a bounce.\n    *   The bands are relatively wide, indicatin
```

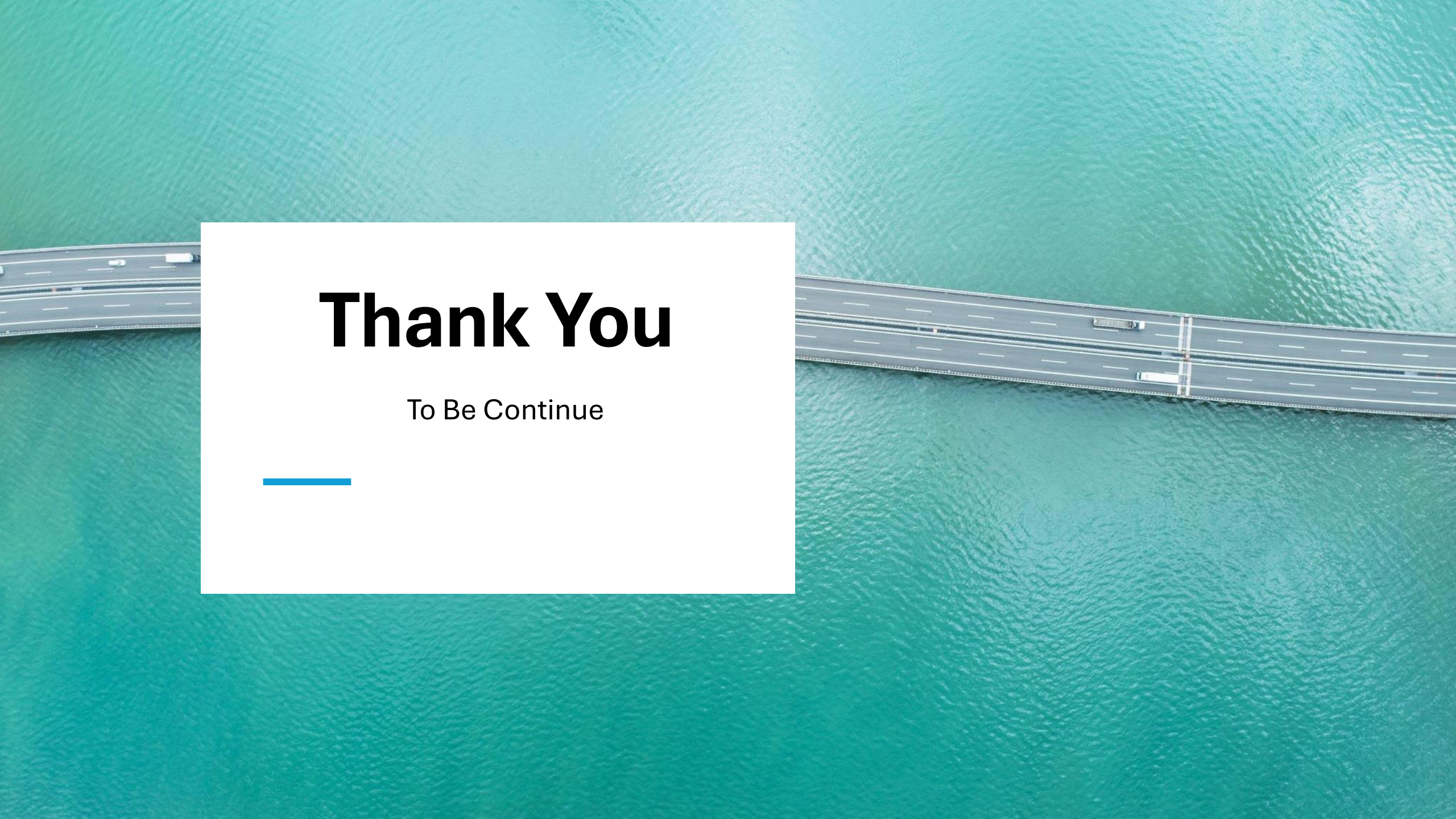
# Agent Final Analyst Report

# FINAL MARKET ANALYST REPORT

FINAL TRANSACTION PROPOSAL: **SELL**

The **ADX** has been declining from previous higher values (e.g., 47.37 on 2025-11-26), suggesting that the strength of the downtrend is starting to wane, but it is still categorized as a strong trend. This could imply that while the bearish trend is still dominant, its intensity is decreasing.

**Overall Market Structure Assessment:** The market structure for BTC as of 2025-12-15 appears to be in a **strong downtrend**, but with some signs of potential short-term exhaustion or a temporary bounce.



# Thank You

To Be Continue